

Atypical Presentation of Acquired Hypoganglionosis in a Patient Taking Clozapine

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Published case report: PMC7333039 / PMID 32670642

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2. Case Report/Case Presentation

A 45-year-old Caucasian male with a history of schizoaffective disorder and chronic constipation was brought to the emergency department for acute encephalopathy. He was somnolent and unresponsive to questions. His home medications included a total of 1,200 mg of clozapine daily. It is unclear how long he was taking this dose. He was also taking risperidone, propranolol, linaclotide, benztropine, alprazolam, and diphenhydramine. The patient's medical team could not determine exactly how many years the patient had been taking clozapine, but family and the patient's medical team indicated that he was started on the clozapine therapy years ago in his adulthood. He had never been diagnosed with ileus or other complications of constipation. He had no surgical history. On arrival, the patient was febrile to 38.5°Celsius and tachycardic but otherwise hemodynamically stable. Initial abdominal examination was soft, nontender, mildly distended with reduced bowel sounds.

Laboratory studies showed a clozapine level of 735 mcg/L and norclozapine level of 449 mcg/L (reference value normal 25–440 mcg/L). Other labs were unremarkable except for a hemoglobin level of 11.8 g/dL. Cerebrospinal fluid, blood cultures, and urine cultures were negative. Computed tomography (CT) of the head without contrast was unremarkable. He was found to have significant urinary retention on bladder ultrasound. Given the negative infectious workup and his clinical presentation, along with his multitude of medications with anticholinergic side effects, a working diagnosis of anticholinergic toxicity was made, and the patient's home clozapine was held.

On hospital day 2, he developed bilious emesis and physical exam revealed a soft, distended, and tympanic abdomen that was mildly tender. Abdominal CT showed a large stool burden diffusely in the small and large intestines without obvious transition points that was concerning for ileus. The patient underwent aggressive medical therapy including frequent tap water enemas and polyethylene glycol administered via nasogastric tube and per rectum. He remained NPO (nulla per os) and underwent intermittent nasogastric tube suctioning. Large amounts of stool were evacuated, but the patient continued to have bilious vomiting. Repeat abdominal CT showed fecalization of stool in the small intestine and dilatation of the proximal and rectosigmoid colon to 8.4 cm and 7 cm, respectively, without a clear transition point. The patient was transferred to the intensive care unit and treated with two doses of intravenous neostigmine with significant

evacuation of stool. Repeat abdominal CT demonstrated bowel without stool but with significantly worsened dilatation. The patient underwent total colectomy with end ileostomy. Given the remarkable dilatation of the rectum, there was concern that primary ileorectal anastomosis would have led to an inadequate functional outcome, and the decision was made to perform an end ileostomy. The patient recovered rapidly to his mental and functional baseline postoperatively with resumption of normal bowel movements.

Given the patient's severe psychiatric symptoms refractory to other medications, a multidisciplinary meeting with psychiatry was held and it was decided to slowly restart the patient on clozapine with monitoring of symptoms and drug levels. Clozapine was titrated slowly with close monitoring for side effects. The repeat serum clozapine and norclozapine levels were 433 mcg/L and 121 mcg/L, respectively. There was no recurrence of systemic clozapine toxicity or symptoms of gastrointestinal dysmotility at these lower levels.

Full-thickness sections from the ileum (Figure 1), cecum (Figure 2), appendix, colon, sigmoid, and rectum (Figures 3 and 4) were obtained and stained with hematoxylin and eosin (H&E), calretinin, and for S-100, a neuronal marker. Final pathologic diagnosis showed hypertrophy of the colonic muscularis propria in the rectum, sigmoid, and cecum. There was focal loss of muscularis propria in the proximal colon. Though there were some ganglia present in the myenteric and submucosal plexuses, they were only focally present in the proximal colon and appendix. Sections from the ileum showed similar muscle hypertrophy and focally absent ganglia in both plexuses consistent with acquired HG.

Figure 1.

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Ileum. (a) Hypertrophy of muscularis propria (magnification 10x, H&E). (b) Focally absent ganglion cells (magnification 20x, S100). (c) Auerbach's plexus highlighted in dark brown (magnification 10x, S-100). (d) Numerous ganglion cells in some areas (magnification 20x, S100).

Figure 2.

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Cecum. (a) Reduced ganglion cells (magnification 40x, H&E). (b) Reduced ganglion cells (magnification 40x, S100).

Figure 3.

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Hypertrophy of the muscularis propria in rectum and sigmoid sections. (a) Rectum (magnification 10x, H&E). (b) Sigmoid colon (magnification 10x, H&E).

Figure 4.

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Sigmoid colon. (a) Focally absent ganglion cells (magnification 40x, H&E). (b) Focally absent ganglion cells (magnification 40x, S100).